

Two-Orbit Packing for Exact State Complexity of Explicit Binary Consensus in Anonymous Dynamic Networks with Periodic Time

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Abstract

We study deterministic binary consensus with explicit termination in anonymous synchronous 1-interval-connected dynamic networks under one-bit broadcast-counting communication. Each round t exposes a free globally aligned phase $\phi_t = t \bmod P$, while the full round number is unavailable. A recent state-phase analysis showed that when a dynamic-diameter bound D is known but the network size is unknown, the minimum persistent-state count $S_D(D, P)$ lies between a lower bound of order D/P and the block-counter upper bound $\lceil D/P \rceil + 2$.

We close this gap exactly. The key is to pack the two opposite homogeneous executions simultaneously. If τ_0 and τ_1 are the first final times on the static cycle C_{2D} , an adaptive split of the cycle implies $\tau_0 + \tau_1 \geq D$. The pre-final phase-augmented trajectories of the all-zero and all-one executions are pairwise disjoint, and their two absorbing final states contribute a full period of P augmented states each. Hence

$$P|Q| \geq \tau_0 + \tau_1 + 2P \geq D + 2P,$$

which matches the known upper construction. Therefore, for every $D, P \geq 1$,

$$S_D(D, P) = \left\lceil \frac{D}{P} \right\rceil + 2.$$

Equivalently, an S -state algorithm exists exactly when $P(S - 2) \geq D$, and the minimum phase period compatible with $S \geq 3$ persistent states is $\lceil D/(S - 2) \rceil$. In particular, the three-state threshold is exactly $P \geq D$.

The same two-orbit argument also sharpens the known-size lower bound to

$$S_n(n, P) \geq 2 + \left\lceil \frac{\lfloor n/2 \rfloor}{P} \right\rceil,$$

while the existing upper bound remains $\lceil (n - 1)/P \rceil + 2$. The claims are restricted to the stated anonymous dynamic-consensus model; no novelty is claimed for generic finite-state pumping or clock-memory tradeoffs.

Keywords. anonymous dynamic networks; binary consensus; explicit termination; state complexity; local memory; periodic clock; dynamic diameter; finite-state lower bounds; time-space tradeoff.

1 Introduction

Explicit termination turns elapsed time into a computational resource. In anonymous dynamic networks, a binary value itself can be stabilized with one bit by monotone flooding, but a process that must irrevocably stop also needs evidence that information had enough time to propagate. Yonezu [1] studied this cost when every node receives the free common phase

$$\phi_t = t \bmod P$$

but not the full round number. For known dynamic diameter D and unknown network size, that work proved

$$\max \left\{ 3, \left\lceil \frac{\lceil D/2 \rceil + 1}{P} \right\rceil \right\} \leq S_D(D, P) \leq \left\lceil \frac{D}{P} \right\rceil + 2,$$

and obtained the asymptotic law $M_D(D, P) = \Theta(\log(2 + D/P))$, where $M_D = \lceil \log_2 S_D \rceil$.

The factor-two gap in the state lower bound came from forcing *one* of the all-zero and all-one homogeneous executions to wait. This note observes that the two executions can be packed together. Two ingredients make the strengthening exact.

First, the split-cycle argument can be made asymmetric. Instead of bisecting C_{2D} in advance, choose the block lengths after the two homogeneous first-final times τ_0, τ_1 are fixed. This yields the sum constraint $\tau_0 + \tau_1 \geq D$.

Second, opposite homogeneous executions cannot share a phase-augmented pre-final state: once they meet at the same pair (q, ϕ) , deterministic homogeneous evolution has the same future, contradicting opposite unanimity outputs. Moreover, each absorbing final state persists while the external phase cycles through all P values. Thus the two final states consume $2P$ augmented pairs in addition to the two disjoint pre-final trajectories.

Main contributions.

1. A two-orbit packing lemma that counts two disjoint homogeneous trajectories together with their absorbing phase cylinders.
2. The exact known-diameter state complexity

$$S_D(D, P) = \lceil D/P \rceil + 2.$$

3. The exact inverse feasibility law $P(S - 2) \geq D$ and, consequently, the exact three-state threshold $P \geq D$.
4. A stronger known-size lower bound

$$S_n(n, P) \geq 2 + \lceil \lfloor n/2 \rfloor / P \rceil,$$

improving the previous cycle bound by roughly a factor of two in the unsaturated regime.

2 Model

We use the model of [1]. The node set V is fixed, nodes are anonymous and deterministic, and time proceeds in synchronous propagation rounds. In round t , the adversary chooses a simple undirected connected graph G_{t+1} . Each node broadcasts one bit to all current neighbors and observes only the numbers (c_0, c_1) of received zero and one broadcasts.

The free external phase is

$$\phi_t = t \bmod P \in \{0, \dots, P-1\},$$

globally aligned and execution-independent. An algorithm has a finite persistent state set Q . Broadcast and transition functions may inspect the phase, but the full round number is otherwise unavailable.

Finality and output are properties of persistent state alone:

$$F \subseteq Q, \quad \text{out} : Q \rightarrow \{0, 1\}.$$

Once a node enters a final state, its persistent state and output never change, although it remains present in communication. The task requires termination, agreement, and unanimity validity.

For completeness, we recall the dynamic-diameter convention used in [1]. For a start time $s \geq 0$ and source node u , let $R_0^s(u) = \{u\}$ and recursively define

$$R_{r+1}^s(u) = R_r^s(u) \cup N_{G_{s+r+1}}(R_r^s(u)),$$

where $N_G(A)$ denotes the set of vertices adjacent in G to at least one vertex of A . A graph sequence has dynamic diameter at most D if $R_D^s(u) = V$ for every start time s and every source u . Thus, on a static connected graph, this notion reduces to ordinary graph diameter.

For known D and unknown n , $S_D(D, P)$ is the minimum $|Q|$ over algorithms that work for every network size and every graph sequence of dynamic diameter at most D . For known n , $S_n(n, P)$ is defined analogously over all n -node 1-interval-connected dynamic networks.

3 Homogeneous augmented orbits

Fix a static cycle C_m and a deterministic period- P algorithm. In the homogeneous execution E_b , all nodes start with input $b \in \{0, 1\}$. By anonymity and cycle symmetry, all nodes have the same persistent state $q_t^{(b)}$ at every time t . Let

$$\tau_b = \min\{t : q_t^{(b)} \in F\}$$

be the first final time, and define the augmented state

$$X_t^{(b)} = (q_t^{(b)}, \phi_t).$$

On a homogeneous cycle there is a deterministic time-homogeneous map

$$G : Q \times \{0, \dots, P-1\} \rightarrow Q \times \{0, \dots, P-1\}$$

with $X_{t+1}^{(b)} = G(X_t^{(b)})$.

Lemma 1 (No pre-final repetition). *For each $b \in \{0, 1\}$, the augmented states*

$$X_0^{(b)}, \dots, X_{\tau_b-1}^{(b)}$$

are pairwise distinct.

Proof. If $X_i^{(b)} = X_j^{(b)}$ for $0 \leq i < j < \tau_b$, deterministic iteration of G repeats the suffix from time i . Since the execution eventually reaches a final state, the same final augmented state would then be reached earlier than τ_b , a contradiction. $\square$

Lemma 2 (Opposite pre-final orbits are disjoint). *The sets*

$$N_b = \{X_t^{(b)} : 0 \leq t < \tau_b\}$$

satisfy $N_0 \cap N_1 = \emptyset$.

Proof. Suppose $X_i^{(0)} = X_j^{(1)}$ for some $i < \tau_0$ and $j < \tau_1$. From that common augmented state onward, both homogeneous executions follow the same deterministic iterates of G . Because the algorithm terminates, both suffixes eventually enter the same persistent final state and therefore have the same final output. This contradicts unanimity validity, which requires output zero in E_0 and output one in E_1 . $\square$

Let $f_b = q_{\tau_b}^{(b)}$ be the persistent final state reached by E_b . Unanimity validity implies $f_0 \neq f_1$. Since final states are persistent, after reaching f_b the execution visits

$$A_b = \{(f_b, \phi) : \phi = 0, \dots, P-1\}$$

as the external phase completes a full cycle. Each A_b has size P .

Lemma 3 (Two-orbit packing). *For homogeneous opposite executions on a static cycle,*

$$P|Q| \geq \tau_0 + \tau_1 + 2P.$$

Proof. The two pre-final sets N_0, N_1 are internally repetition-free and mutually disjoint. Each has size τ_b . The final phase cylinders A_0, A_1 each have size P and are disjoint because $f_0 \neq f_1$. They are also disjoint from $N_0 \cup N_1$, since every state appearing in a pre-final set is nonfinal while $f_0, f_1 \in F$. Hence $Q \times \{0, \dots, P-1\}$ contains at least

$$|N_0| + |N_1| + |A_0| + |A_1| = \tau_0 + \tau_1 + 2P$$

distinct augmented pairs. $\square$

4 A sum-waiting lemma on split cycles

We make explicit the locality fact used below; it is the phase-aware version of the standard synchronous radius bound and is also used in [1].

Lemma 4 (Phase-aware radius locality). *Fix P and a deterministic period- P algorithm on a static graph. After $t \geq 0$ propagation rounds, the persistent state of a node v is determined by the initial states within graph distance at most t from v , together with the common phase prefix $\phi_0, \dots, \phi_{t-1}$. Consequently, if two executions have identical radius- t initial neighborhoods around their respective distinguished nodes, then those nodes have identical persistent states at time t .*

Proof. Induct on t . At $t = 0$, the persistent state is determined by the node's own initial state. Suppose the claim holds after t rounds. In round t , the next state of v is a deterministic function of its current state, the common phase ϕ_t , and the multiset counts of the one-bit messages sent by its neighbors. By the induction hypothesis, each neighbor's current state, and hence its round- t broadcast, is determined by initial states within distance at most t of that neighbor. Their union lies within distance at most $t+1$ of v . Therefore the state of v at time $t+1$ is determined by its radius- $(t+1)$ initial neighborhood and the phase prefix through ϕ_t . $\square$

Lemma 5 (Adaptive split on C_{2D}). *For $D \geq 2$, let τ_0, τ_1 be the first final times of the two homogeneous executions on C_{2D} . Then*

$$\tau_0 + \tau_1 \geq D.$$

Proof. Assume $\tau_0 + \tau_1 \leq D - 1$. Give a contiguous block of

$$L_0 = 2\tau_0 + 1$$

cycle vertices input zero and give all remaining vertices input one. The one-block length is

$$L_1 = 2D - L_0 \geq 2\tau_1 + 1.$$

Choose a center u of the zero block. Its distance from the one block is $\tau_0 + 1$, so its radius- τ_0 initial neighborhood is all zero. Choose a deepest vertex v of the one block; its distance from the zero block is at least $\tau_1 + 1$, so its radius- τ_1 initial neighborhood is all one.

By the phase-aware radius-locality lemma, u follows its all-zero homogeneous history through time τ_0 and therefore enters an immutable final state with output zero. Likewise, v enters an immutable final state with output one by time τ_1 . The two decisions coexist in the same execution, contradicting agreement. $\square$

Lemma 6 (The case $D = 1$). *On K_2 , the homogeneous first-final times satisfy $\tau_0 + \tau_1 \geq 1$.*

Proof. If $\tau_0 = \tau_1 = 0$, then both input values initialize directly into their respective immutable final states. On the mixed-input K_2 , the two nodes therefore begin with opposite immutable outputs, violating agreement. $\square$

5 Exact state complexity for known dynamic diameter

Theorem 1 (Exact known-diameter state complexity). *For every $D \geq 1$ and $P \geq 1$,*

$$S_D(D, P) = \left\lceil \frac{D}{P} \right\rceil + 2.$$

Proof. For the lower bound, use C_{2D} when $D \geq 2$ and K_2 when $D = 1$. These are admissible static networks of dynamic diameter exactly D . The sum-waiting lemmas give

$$\tau_0 + \tau_1 \geq D.$$

Two-orbit packing therefore yields

$$P|Q| \geq D + 2P,$$

so

$$|Q| \geq 2 + \left\lceil \frac{D}{P} \right\rceil.$$

For the upper bound, use the block-counter algorithm of [1]. A zero-input node immediately enters an absorbing final-zero state and continues broadcasting zero. A one-input node remains live while storing only the number of completed phase periods. Dynamic diameter at most D guarantees that an initial zero reaches every node within D propagation rounds. With $B = \lceil D/P \rceil$, the persistent states are

$$\{\text{final0}, L_0, \dots, L_{B-1}, \text{final1}\},$$

for a total of $B + 2 = \lceil D/P \rceil + 2$ states. On the last propagation round, hearing zero has priority over timing out to final one. Thus the lower and upper bounds match. $\square$

Corollary 1 (Exact persistent-bit complexity). *For every $D, P \geq 1$,*

$$M_D(D, P) = \left\lceil \log_2 \left(\left\lceil \frac{D}{P} \right\rceil + 2 \right) \right\rceil.$$

Corollary 2 (Exact feasible state-phase region). *Fix $D \geq 1$ and an integer state budget $S \geq 3$. There exists a correct algorithm with at most S persistent states if and only if*

$$\boxed{P(S - 2) \geq D.}$$

Equivalently, the minimum phase period permitting an S -state solution is

$$\boxed{P_{\min}(D, S) = \left\lceil \frac{D}{S - 2} \right\rceil.}$$

Proof. By the exact theorem, $S_D(D, P) \leq S$ iff $\lceil D/P \rceil + 2 \leq S$. Since $S - 2$ is an integer, this is equivalent to $D/P \leq S - 2$, hence to $D \leq P(S - 2)$. $\square$

Corollary 3 (Exact three-state threshold). *For every $D \geq 1$,*

$$S_D(D, P) = 3 \iff P \geq D.$$

6 A sharper known-size lower bound

The same packing principle strengthens the known-size result even though it does not close the remaining gap.

Lemma 7 (Adaptive split on C_n). *For $n \geq 4$, let τ_0, τ_1 be the homogeneous first-final times on C_n . Then*

$$\tau_0 + \tau_1 \geq \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. Assume instead that

$$\tau_0 + \tau_1 \leq \left\lfloor \frac{n}{2} \right\rfloor - 1.$$

Assign zero to a contiguous block of $L_0 = 2\tau_0 + 1$ vertices and one to the remaining $L_1 = n - L_0$ vertices. The assumed inequality gives

$$L_1 \geq 2\tau_1 + 1.$$

Each block therefore contains a vertex whose radius- τ_b neighborhood is homogeneous in value b . The phase-aware radius-locality lemma forces those two vertices to reproduce opposite homogeneous final decisions, contradicting agreement. $\square$

Theorem 2 (Improved known-size state bounds). *For every $n \geq 4$ and $P \geq 1$,*

$$\boxed{2 + \left\lceil \frac{\lfloor n/2 \rfloor}{P} \right\rceil \leq S_n(n, P) \leq 2 + \left\lceil \frac{n - 1}{P} \right\rceil.}$$

Proof. The lower bound follows from the adaptive split lemma and two-orbit packing on C_n :

$$P|Q| \geq \left\lfloor \frac{n}{2} \right\rfloor + 2P.$$

The upper bound is the known monotone-flooding block counter with horizon $n - 1$ [1]. $\square$

Remark 1. At $P = 1$, the new lower bound is $S_n(n, 1) \geq \lfloor n/2 \rfloor + 2$, compared with the previous split-cycle lower bound of order $n/4$. Determining the exact known-size state complexity remains open in this note.

7 Why the extra $2P$ is essential

The exact known- D theorem is not obtained merely by replacing a maximum waiting time with a sum. The persistent state count has two irreducible absorbing-state costs. In augmented state space, each final persistent state sweeps through all P phase values after termination. Thus the two opposite unanimity outputs reserve $2P$ augmented pairs. The remaining $|Q| - 2$ persistent states can account for at most $P(|Q| - 2)$ distinct pre-final phase/state pairs. The diameter witness forces at least D such pre-final pairs across the two opposite homogeneous executions. This is precisely the inequality

$$P(|Q| - 2) \geq D.$$

The upper construction meets it with equality at the level of integer ceilings.

This interpretation also explains the inverse law. The two final states are a fixed state tax imposed by two-output explicit termination. Every additional persistent state supplies exactly one more period, or P rounds, of distinguishable waiting capacity in the optimal diameter-aware construction.

8 Relation to prior work

The direct predecessor is the state-phase tradeoff analysis in [1], which introduced the free modulo- P phase model for this task, proved the block-counter upper bound, and established asymptotically matching lower bounds. The present note changes the lower-bound accounting: it couples the two opposite homogeneous executions and counts their absorbing phase cylinders, yielding an exact result for known D and a stronger known- n bound.

Parzych and Daymude [3] prove logarithmic memory lower bounds for terminating anonymous dynamic broadcast, establishing that termination memory is a genuine resource in nearby models. Turau [4] gives a randomized broadcast algorithm with stabilizing termination using $O(\log \log n)$ bits per node in anonymous synchronous 1-interval-connected dynamic networks; the task, randomization, start semantics, and asymptotic memory objective differ from the deterministic exact consensus-state question here. Blanc, Di Luna, and Viglietta [2] show that rich computation remains possible under one-bit broadcast-counting communication. Di Luna and Viglietta [5] study finite-state and self-stabilizing computation in anonymous dynamic networks.

The broad idea of combining a short clock with local storage is older and is not claimed here. Lenzen et al. [6] explicitly combine a synchronized short clock with local round-label storage in a fault-tolerant global-time construction. Charron-Bost and Penet de Monterno study modulo-clock synchronization in dynamic networks [7] and later bounded-memory lower bounds for clock synchronization under arbitrary starts [8]. Their task is to create or synchronize the clock itself, whereas the present model supplies the globally aligned phase $t \bmod P$ for free and measures the additional persistent state needed for explicitly terminating binary consensus. Bazzi et al. [9] construct a deterministic synchronizer for anonymous networks under arbitrary dynamics using an extended Pull model with a one-bit multi-writer atomic register at each edge-port; this is a different communication model and synchronization objective. Aspnes [10] studies clocked population protocols, and Mayo and Kearns [11] study termination detection with roughly synchronized physical clocks.

A targeted literature search through September 5, 2026 did not identify a prior theorem giving the exact formula $S_D(D, P) = \lceil D/P \rceil + 2$, or the equivalent feasibility law $P(S - 2) \geq D$, for the stated anonymous dynamic binary-consensus model. This is a scoped literature-screening statement, not a certification of novelty.

9 Claim boundary and limitations

The paper does not claim novelty for finite-state pumping, disjoint deterministic trajectories in general, static-cycle locality, monotone flooding, dynamic-diameter definitions, periodic clocks, or generic time-space tradeoffs. The claimed contribution is the task-specific combination that yields the exact persistent-state characterization above.

The theorem assumes deterministic anonymous synchronous leaderless 1-interval-connected dynamic networks, a free globally aligned period- P phase, finality and output encoded in persistent state rather than phase, final nodes remaining present in communication, and the one-bit broadcast-counting upper-bound model. Randomization, identifiers, leaders, Byzantine faults, asynchronous communication, and alternative disappearance-on-finalization semantics are outside scope.

The exact lower bound itself uses only static cycles (and K_2 for $D = 1$), so it does not rely on a time-varying adversarial topology schedule or on the one-bit message restriction. The one-bit restriction is relevant to the simple matching upper construction.

10 Conclusion

Opposite unanimous executions contain more state-complexity information than either trajectory alone. By adapting the split witness to their two first-final times and packing both augmented orbits together with the two absorbing phase cylinders, the known-diameter state-phase tradeoff becomes exact:

$$S_D(D, P) = \left\lceil \frac{D}{P} \right\rceil + 2.$$

Equivalently, the exact feasible region is $P(S - 2) \geq D$. The three-state regime therefore begins exactly at $P = D$, not merely by a sufficient upper-bound construction.

The same argument improves the known-size lower bound to $2 + \lceil [n/2]/P \rceil$. Closing the remaining known-size gap is the natural residual problem.

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